

Assignment 7

I did not use any form of AI writing this assignment.

Podcast 11 - Ruth Lehmann and The Saturday Afternoon Experiment

In the podcast Ruth Lehmann introduced herself as not only a Biology Professor but also as a developmental and cell biologist, as well as a geneticist. She is the head director of MIT's Whitehead Institute - a research center which specializes in fundamental biology and biomedical science that aims to improve human health. At the time of the interview she conducted research on germ cells and embryogenesis.

Ruth raised two main points regarding creativity in science. At the beginning of the podcast she stated that in her opinion creativity brings fun into science and that it is the first and most important aspect of scientific research. She believes that creativeness is based on two things - observation and listening. One, that observation makes you question uncertainties and evokes your curiosity to explore something further. In my opinion it is really important to highlight what Ruth has also elaborated on - that observing is not about sitting idly and waiting for an idea to come to you, but about actually getting involved in the research. And two, that listening impels you to actively interact with other scientific fields for you to expand your knowledge and find new connections that otherwise would not surface in the just analysis of your own.

Ruth also discussed how in her opinion the key to creativity can be found in, what she calls, "The Saturday Afternoon Experiments". Those experiments are meant to explore things beyond the original thesis, be the ones to satisfy researcher's curiosity and maybe lead to new discoveries. She believed that that is the time to let yourself be creative and feel the thrill of excitement. However, she also cautioned against the burnout from inevitable failures that may come from "playing around" with spontaneous research. I think that while there are definitive benefits of this approach it is really important to keep in mind that failing is also a part of the research.

At the same time she puts forward a brief thought about how curiosity is something to be opened in young researchers rather than something that is to be learnt. While I definitely see her point I would argue that something more hides behind that statement. In modern times the society has developed a cultural construct of how the only success that counts is that one with the lack of any failure, that disproving a thesis is just a wasted time of conducted research. Such an approach stifles the creativity that humans naturally have from the very beginning. The stereotypical view of science which only focuses on strict guidelines also doesn't encourage scientists to go beyond the given thesis and search the unknown. So while I would emphasise that creativity is something young researchers have to rediscover in themselves, I would also say that they need to learn how and when to give themselves the freedom to just be creative.

To conclude I believe the takeaway message from the podcast with Ruth Lehmann is to always have the attitude that each day is the day of opportunity so we need to stay open and curious to research the science "puzzles" that may lead us to great new discoveries.